



GloFAS v4: towards hyper-resolution hydrological modelling at global scale

Summary: the Global Flood Awareness System (GloFAS) is the **freely available, operational** flood forecasting service of the Copernicus Emergency Management Service.

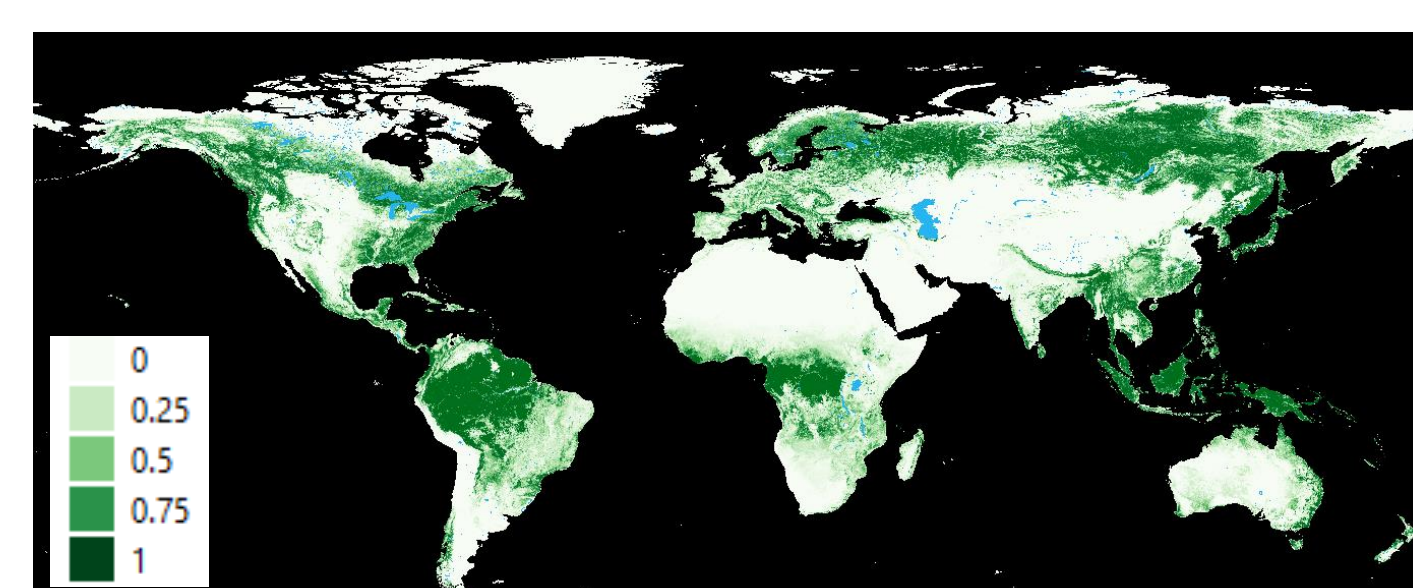
The GloFAS dataset includes medium-range and seasonal discharge forecasts, as well as storages (e.g. soil moisture, snow water equivalent) and main fluxes (e.g. surface runoff).

The currently operational 0.1 degrees (~10km) GloFAS v3 is undergoing a major upgrade which includes (a) the use of entirely revamped input maps with **0.05 degree (~5km)** resolution; (b) an upgraded version of the LISFLOOD-Open Source hydrological model; (c) a combined calibration approach that can benefit of a larger number of observed data.

(a) 0.05 degree resolution input maps

~90 maps providing information on catchments' morphology, land cover and use, soil properties, lakes, reservoirs, water demand for human use

HIGH RESOLUTION and the use of the latest RESEARCH FINDINGS, REMOTE SENSING, and IN SITU DATA allow the accurate representation of the spatial variability of catchments' properties.



Map: fraction of each pixel occupied by forest(-)
Main sources: Copernicus Global Land Cover Layers: CGLS-LC100²; CORINE Land Cover 2018 CLC2018³

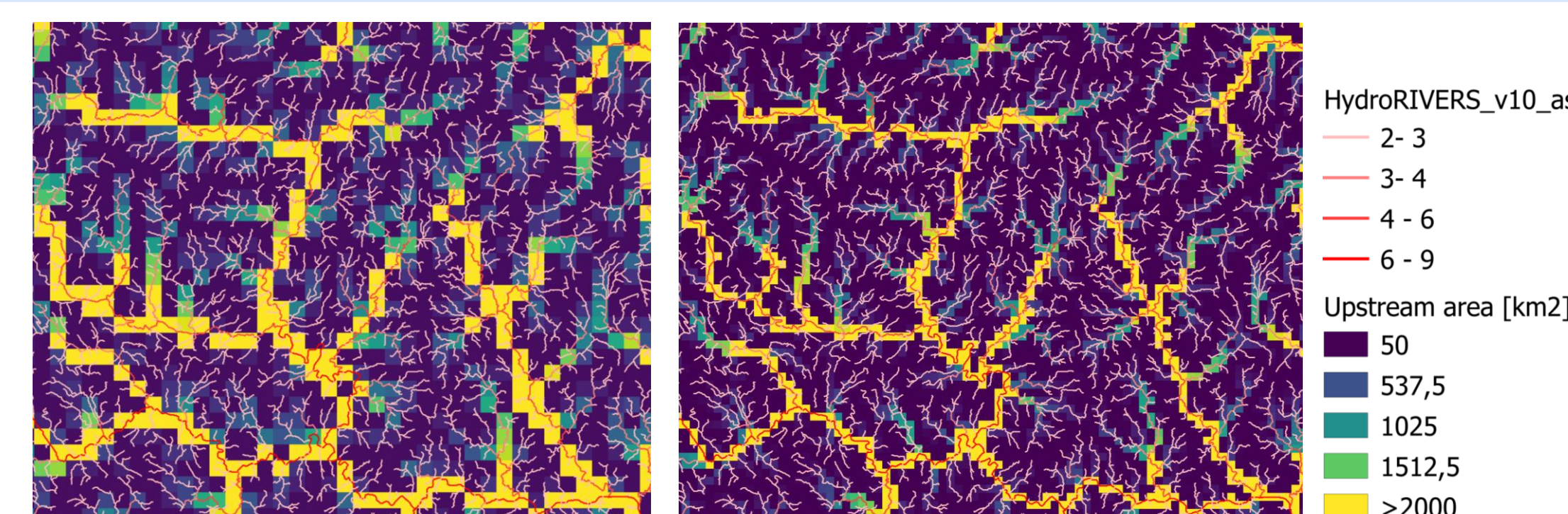
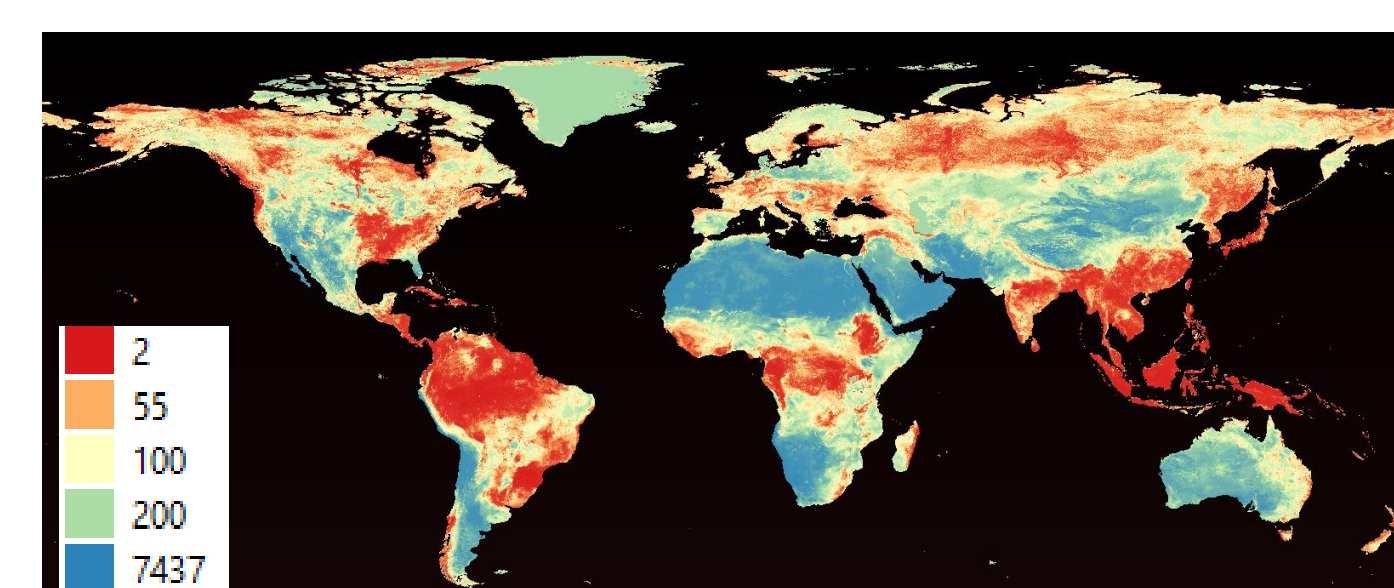


Fig.1: River network at 0.1 degrees (left) and 0.05 degrees resolution (right): drainage upstream area, HydroRIVERS¹ shown as reference.



Map: soil hydraulic conductivity (mm/day)
Main sources: International Soil Reference and Information Centre (ISRIC) SoilGridsTM (2020)⁴; Toth et al. (2015)⁵

Fig.2: Examples of maps and main sources of information. Complete documentation: <https://ec-jrc.github.io/lisflood-code/4-Static-Maps-introduction/>

(b) LISFLOOD-Open Source hydrological model

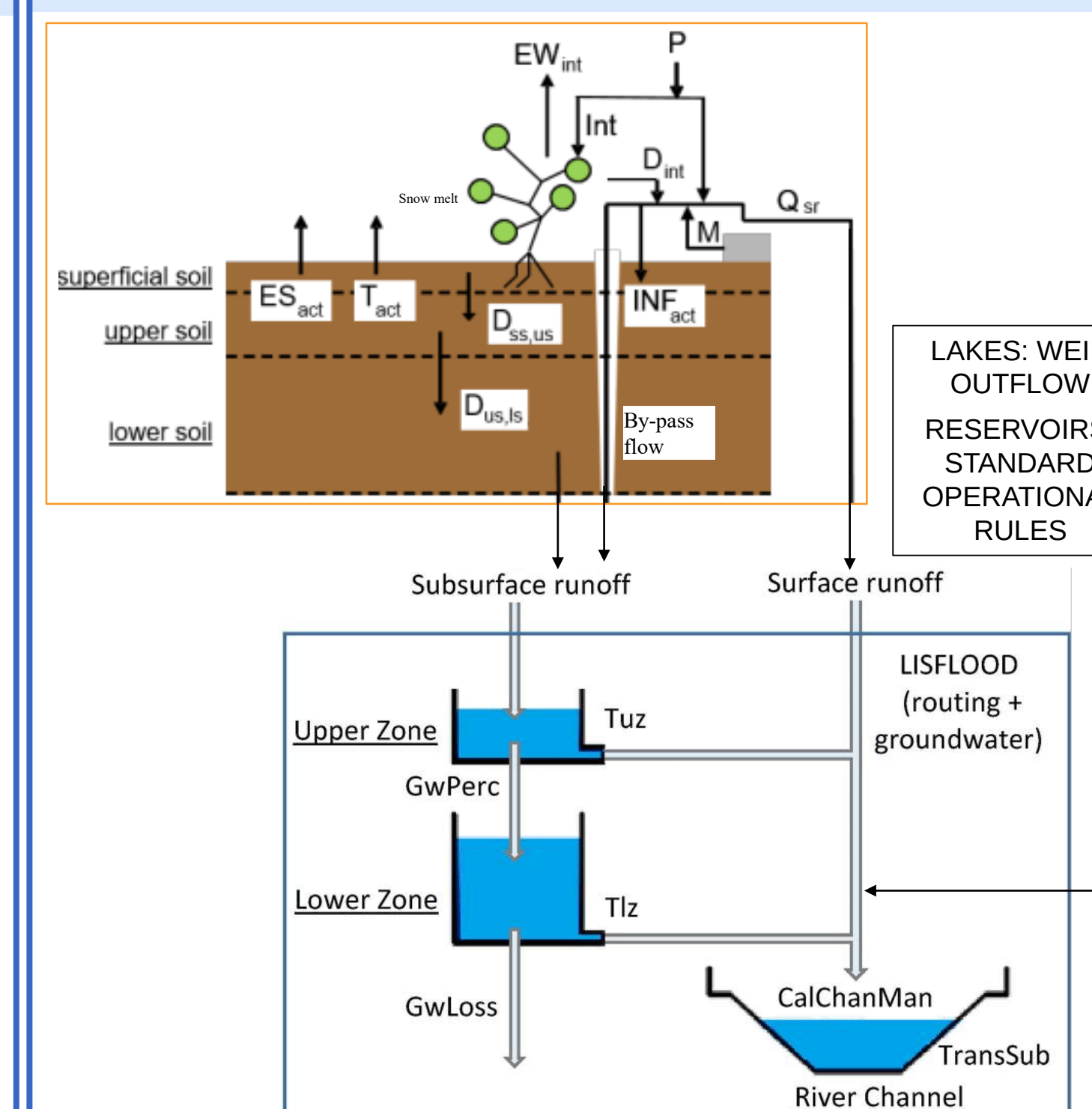


Fig. 3: LISFLOOD hydrological processes, documentation: <https://ec-jrc.github.io/lisflood-model/>

Open Source: <https://github.com/ec-jrc/lisflood-code>

Semi-distributed, physically based model:

- 6 land cover fractions within a cell.
- 3 soil layers; 2 groundwater storages.
- Kinematic wave routing
- Lakes and dams.
- Water abstraction for human use.

Computational efficiency:

- **Optimized management of large input datasets** to make use of high resolution maps;
- Parallel computations.

(c) Model calibration

In situ discharge measurements: top-down approach: Distributed Evolutionary Algorithm for Python⁶, objective function: 'KGE'.

Regionalization: parameters transfer from a donor catchment, the latter selected according to spatial proximity and climatic similarity.

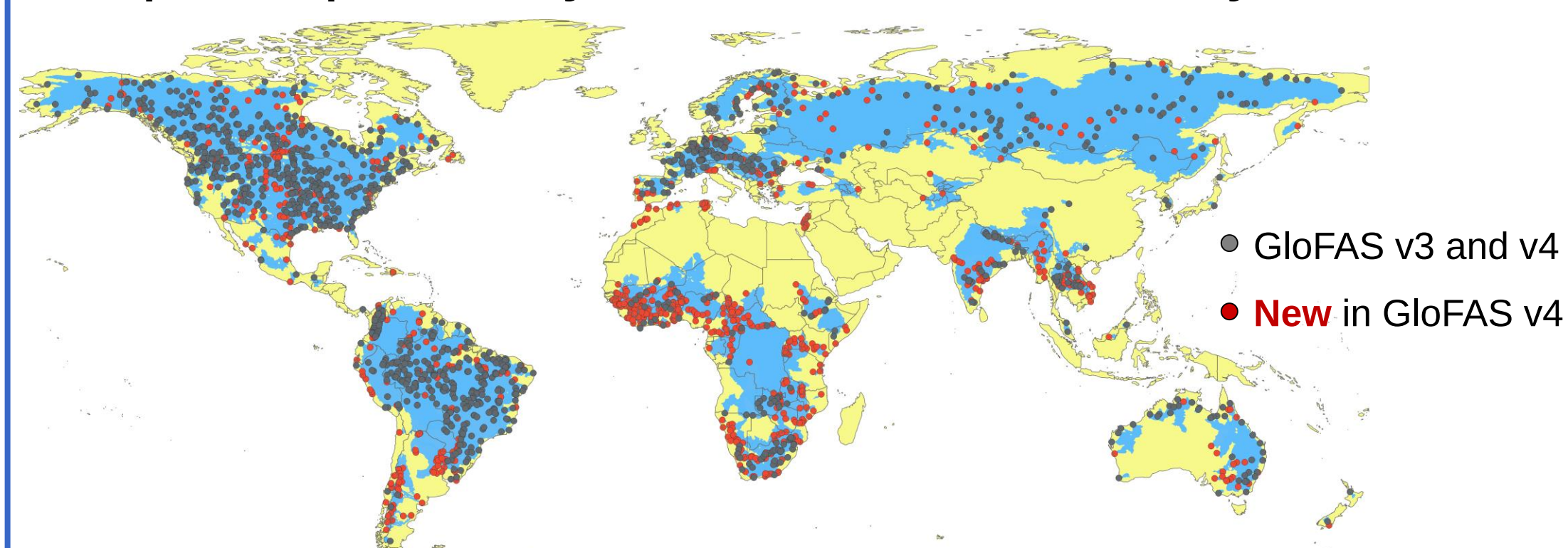


Fig.4: 1996 discharge stations used in calibration, in blue their upstream area

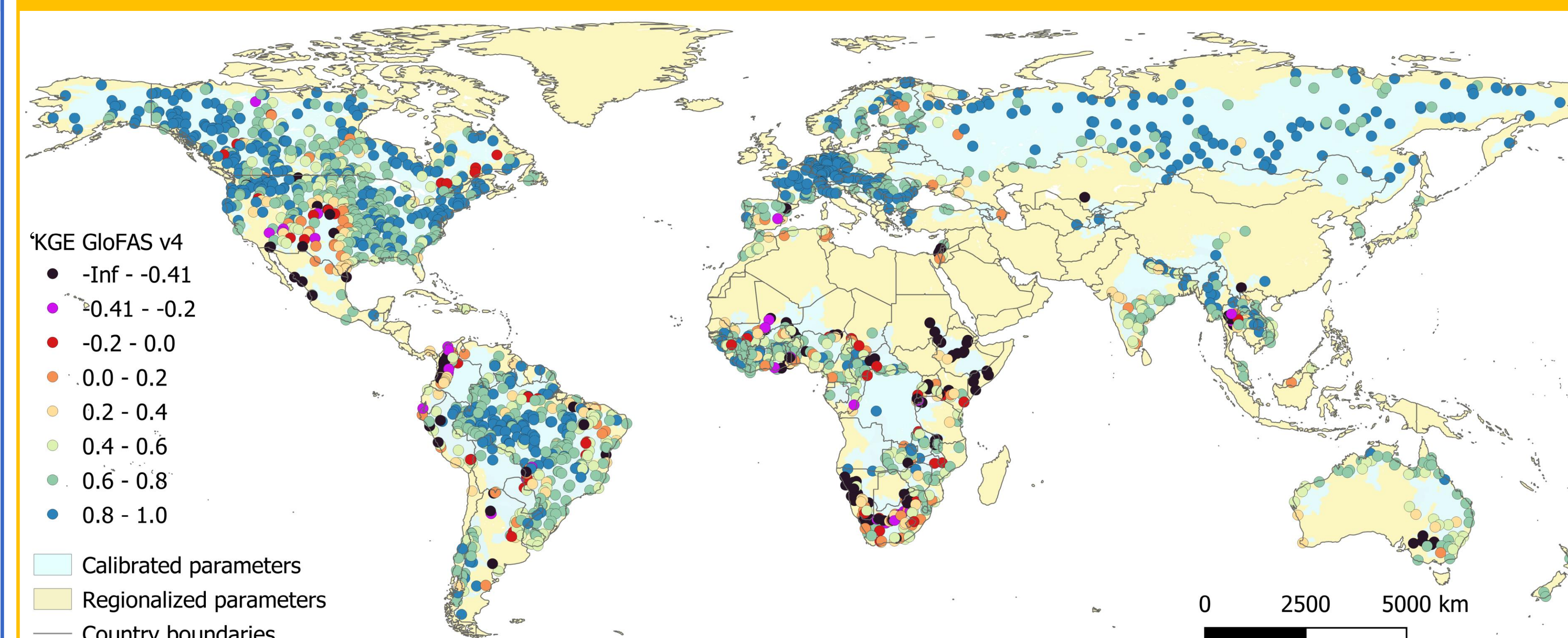


Fig. 5: KGEs values for 1996 calibrated stations

GloFAS v4

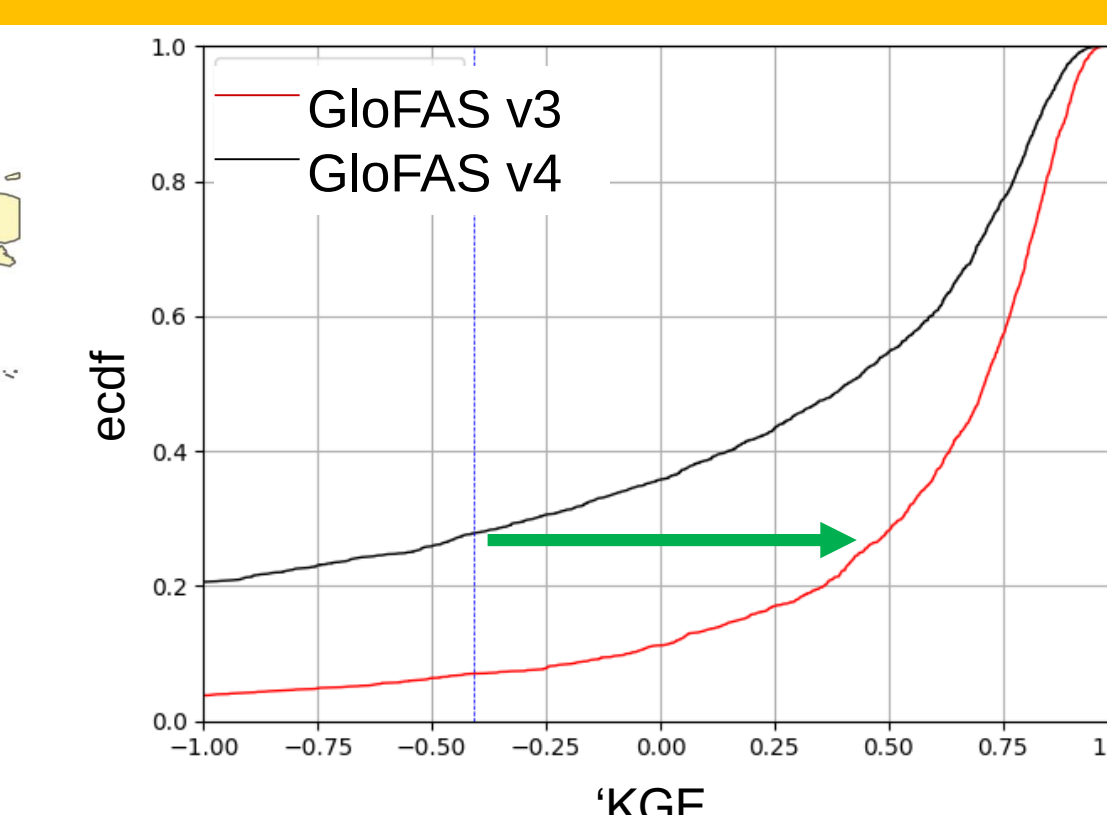


Fig. 6: GloFAS v3 and v4 comparison

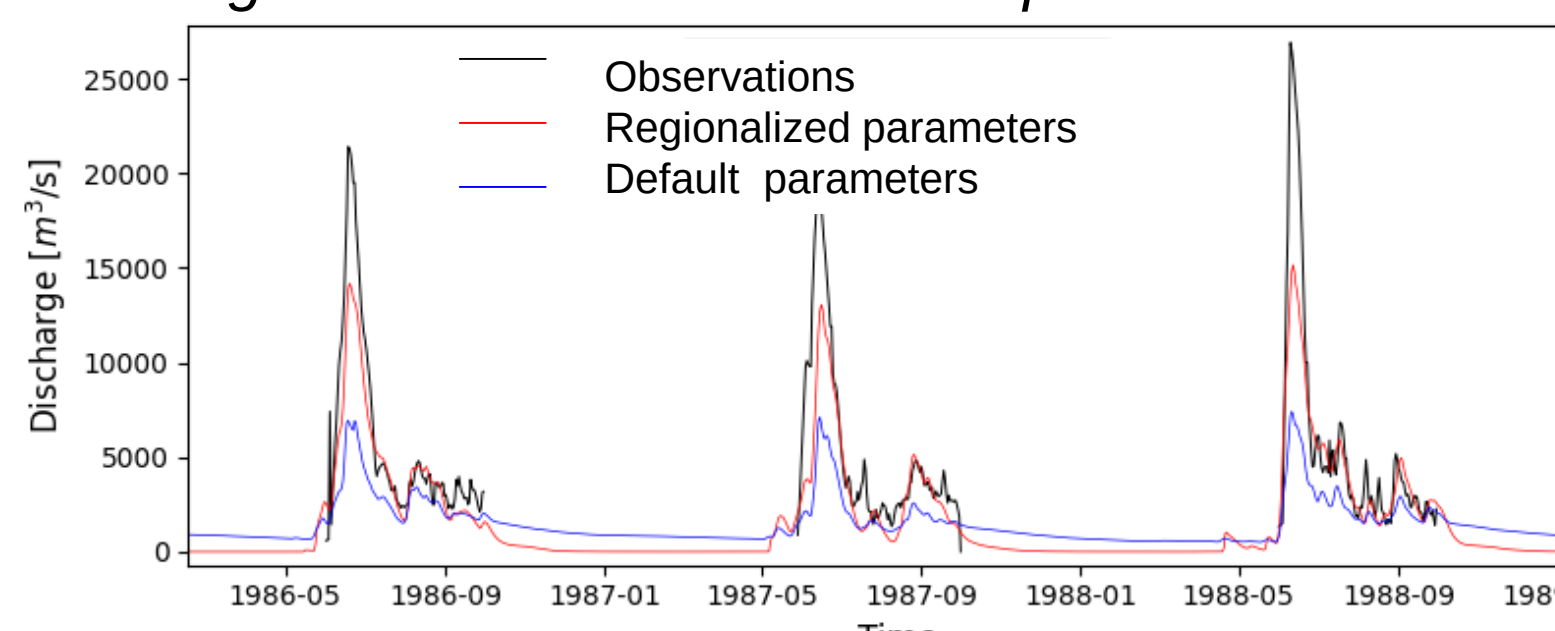


Fig. 7: Example of results in a not-calibrated catchment

The cumulative distribution function of 'KGE values for the 1996 calibration points show the increased accuracy of the GloFAS v4 set-up compared to the lower resolution v3 set-up.

The regionalized parameters were tested using gauge stations not included in calibration.

Concluding remarks: the new high resolution set-up, and the new calibration of the hydrological model will lead to more timely and accurate flood forecasts.

The operational release of GloFAS v4 is foreseen at the end of July 2023. LISFLOOD, the high-resolution implementation maps, and GloFAS v4 reanalysis dataset are (will soon be) publicly available and they disclose opportunities for further analysis of the terrestrial water cycle fluxes and storages, and of the current and future state of global water resources.

References:

¹<https://www.hydrosheds.org/products/hydrorivers>

²<https://www.mdpi.com/2072-4292/12/6/1044>

³<https://land.copernicus.eu/pan-european/corine-land-cover/clc2018>

⁴<https://www.isric.org/>

⁵<https://doi.org/10.1111/ejss.12192>

⁶<https://jmlr.org/papers/volume13/fortin12a/fortin12a.pdf>

